■ AUTONOMOUS SMART GRID CONTROLLER

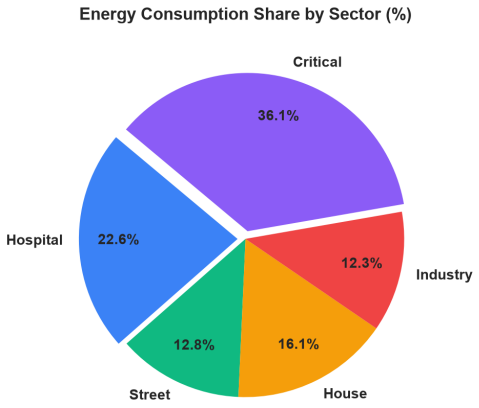
Operational Energy Analytics & IEEE Reliability Engineering Report

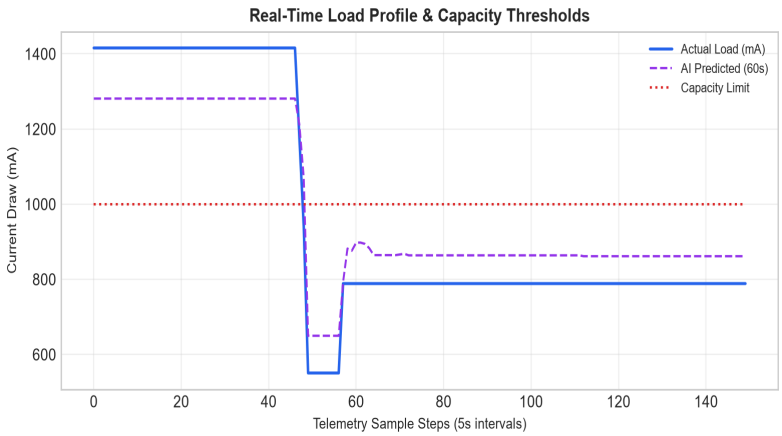
Total Grid Energy: 0.05520 kWh	**Estimated Energy Cost:** ■ 0.44	**Peak Grid Draw:** 3288.0 mA	**Grid Load Factor:** 25.5 %
Autonomy Rate: 64.8%	**SAIFI (Freq Index):** 47.00 /cust	**SAIDI (Duration):** 117.50 min	**AI Curtailed Energy:** 0.04262 kWh

1. Sector-by-Sector Energy & Cost Breakdown

Sector / Branch	Energy (kWh)	Share (%)	Estimated Cost (■)	Safety Limit
Hospital (Priority 4 - Critical)	0.012487	22.6 %	■ 0.0999	272 mA
Street Lighting (Priority 3)	0.007050	12.8 %	■ 0.0564	405 mA
Residential House (Priority 2)	0.008912	16.1 %	■ 0.0713	256 mA
Industrial Load (Priority 1 - Shed First)	0.006801	12.3 %	■ 0.0544	264 mA
Critical Infrastructure (Non-Sheddable)	0.019948	36.1 %	■ 0.1596	320 mA

2. Load Profiling & Power Stability Charts





3. Recent Actuation & Self-Healing Audit Trail

Timestamp	Action	Branch	Reason	Current / Limit
12:19:10	**RESTORE**	HOUSE	HEADROOM_RESTORE	791.5 / 1000 mA
12:18:54	**SHED**	STREET	OVERLOAD_85	391.0 / 1000 mA
12:18:52	**SHED**	HOUSE	OVERLOAD_85	238.0 / 1000 mA
12:18:50	**SHED**	INDUSTRY	OVERLOAD_85	236.0 / 1000 mA
12:10:46	**RESTORE**	HOUSE	HEADROOM_RESTORE	791.5 / 1000 mA
12:10:30	**SHED**	STREET	OVERLOAD_85	391.0 / 1000 mA

12:10:28	**SHED**	HOUSE	OVERLOAD_85	238.0 / 1000 mA
12:10:26	**SHED**	INDUSTRY	OVERLOAD_85	236.0 / 1000 mA